



인하공업전문대학
INHA TECHNICAL COLLEGE

무선 센서 네트워크 12주차~13주차

인하공업전문대학 컴퓨터 정보과
김한결 강사

- 요약

- I. Zigbee TinyOS 기반 온습도 센서 프로그래밍
- II. NesC기반 미들웨어
- III. 라즈베리파이 기반 TinyOS 환경 구축
 - I. 사전 설치
 - II. NesC 설치
 - III. TinyOS Tool 설치

- 온습도 센서 TinyOS 기반 NesC 구현
- Cygwin 실행
 - mkdir 학번 (디렉토리생성)



```
PC@4-405-~  
$ mkdir 2020020
```

- cd 학번 (이동)

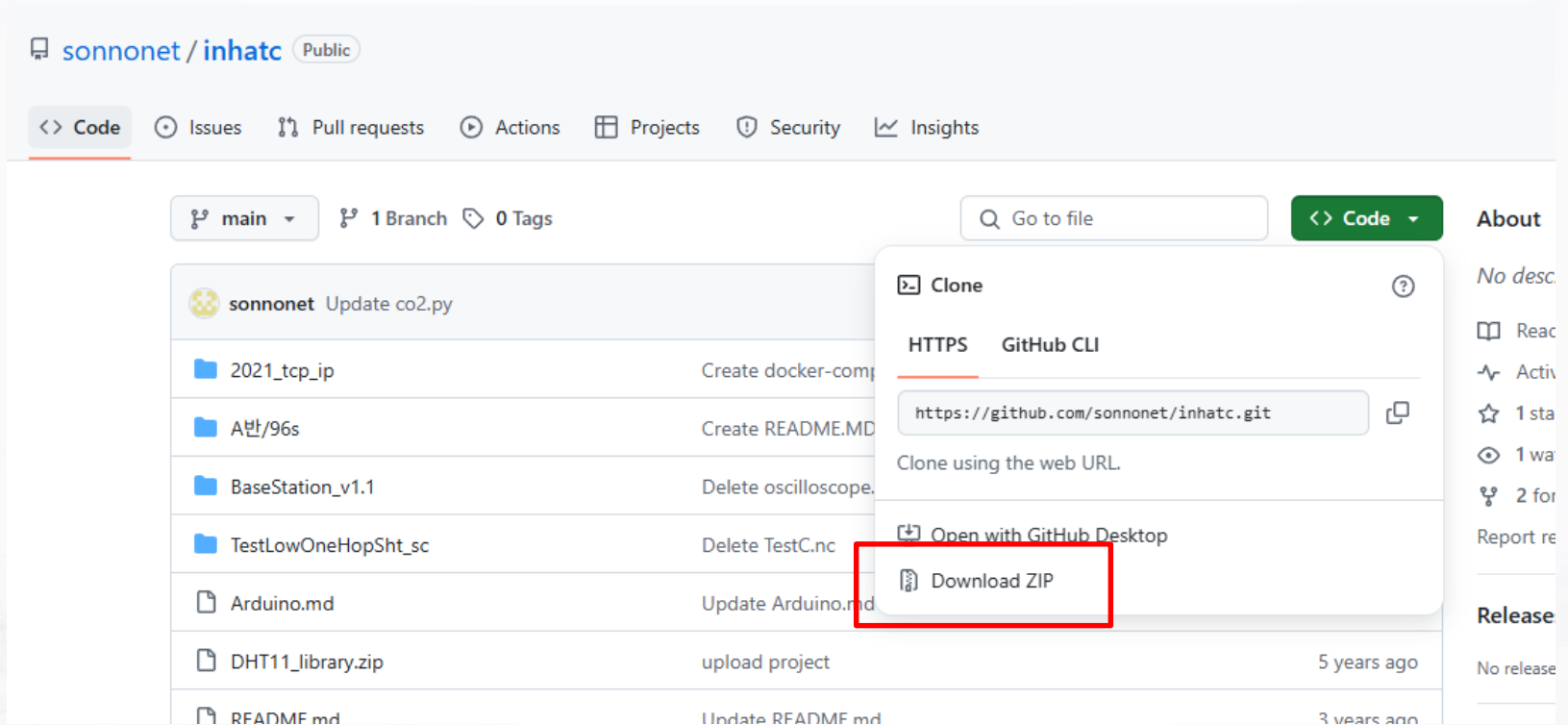


```
PC@4-405-~  
$ cd 2020020/  
  
PC@4-405-~/2020020  
$
```

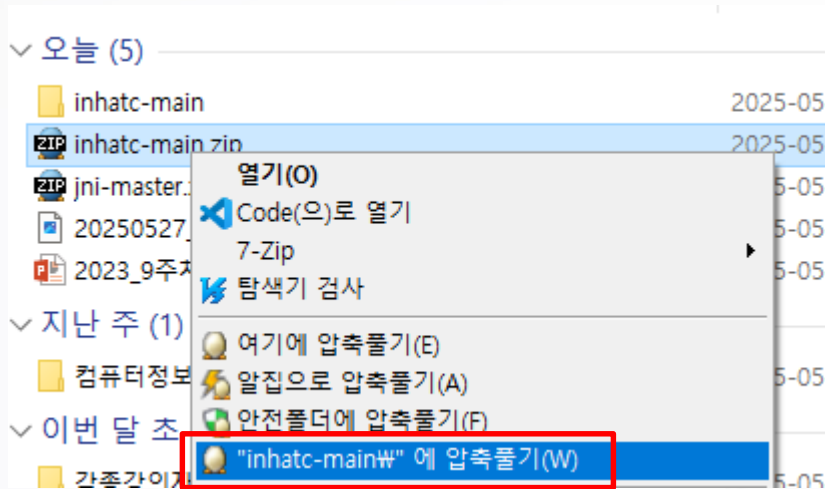
- Github 소스코드 다운
- <https://github.com/sonnonet/inhatc>

	sonnonet Delete TestC.nc	f1be3ec 1 minute ago	🕒 43 commits
	2021_tcp_ip	Create docker-compose.yml	2 years ago
	A반/96s	Create README.MD	2 years ago
	BaseStation_v1.1	Delete oscilloscope.py	7 minutes ago
	TestLowOneHopSht_sc	Delete TestC.nc	1 minute ago

- Github 소스코드 다운



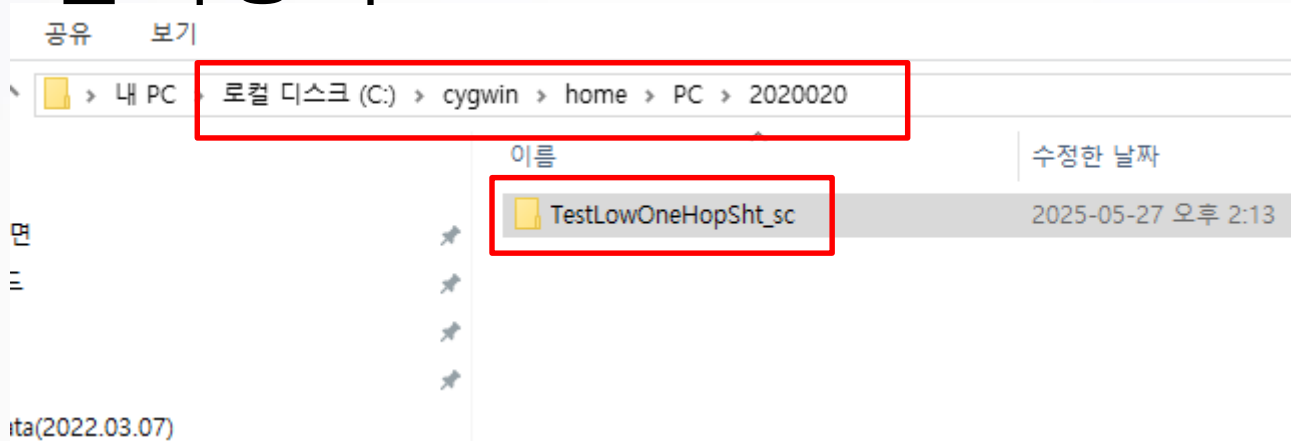
- 압축풀기



- TestLowOneHopSht_sc 복사

이름	수정한 날짜	유형	크기
2021_tcp_ip	2025-05-27 오전 10:18	파일 폴더	
A반	2025-05-27 오전 10:18	파일 폴더	
BaseStation_v1.1	2025-05-27 오전 10:18	파일 폴더	
TestLowOneHopSht_sc	2025-05-27 오전 10:18	파일 폴더	
Arduino.md	2022-12-18 오후 2:59	Markdown 원본 ...	1KB
co2.py	2022-12-18 오후 2:59	PY 파일	1KB
...	...	...	...

- cygwin -> home -> PC(유저이름)->학번
- 붙여넣기



- cygwin 돌아옴
- cd TestLowOneHopSht_sc/

```
PC@4-405-~ /2020020
$ ls
TestLowOneHopSht_sc/

PC@4-405-~ /2020020
$ cd TestLowOneHopSht_sc/

PC@4-405-~ /2020020/TestLowOneHopSht_sc
$ vim TestAppC.nc
```

TinyOS 기반 NesC 프로그래밍 (온습도센서)

- vim TestAppC.nc

```
~/sonnonet_tinyos/Tinyos/TestLowOneHopSht_sc
1 includes Test;
2 configuration TestAppC
3 {
4 }
5 implementation
6 {
7   components TestC, MainC;
8   components LedsC, new TimerMilliC();
9
10  components ActiveMessageC as AMC;
11  components new AMSenderC(AM_TEST_DATA_MSG) as AMSC;
12
13  TestC.Boot -> MainC;
14  TestC.Leds -> LedsC;
15  TestC.MilliTimer -> TimerMilliC;
16
17  TestC.RadioControl -> AMC;
18  TestC.RadioSend -> AMSC;
19
20  components new SensirionSht11C() as Sht11Ch0C;
21  TestC.Temp -> Sht11Ch0C.Temperature;
22  TestC.Humi -> Sht11Ch0C.Humidity;
23
24  components new I1luAdcC() as I1lu;
25  TestC.I1lu -> I1lu;
26
27  components BatteryC;
28  TestC.Battery -> BatteryC;
29 }
```

TestAppC.nc

TinyOS 기반 NesC 프로그래밍 (온습도센서)

- vim Test.h

```
~/sonnonet_tinyos/Tinyos/TestLowOneHopSht_sc
1 #ifndef TEST_H
2 #define TEST_H
3 #include "message.h"
4 enum {
5     TEST_PERIOD = 10240LU,
6 };
7 enum {
8     DFLT_VAL = 0x11,
9 };
10 enum {
11     TEST_DATA_LENGTH = TOSH_DATA_LENGTH - 6,
12 };
13 enum {
14     AM_TEST_DATA_MSG = 0xA4,
15 };
16
17 typedef nx_struct test_data_msg {
18     nx_am_addr_t srcID;
19     nx_uint32_t seqNo;
20     nx_uint16_t type;
21     nx_uint16_t Temp;
22     nx_uint16_t Humi;
23     nx_uint16_t Illu;
24     nx_uint16_t battery;
25     //nx_uint8_t testData[TEST_DATA_LENGTH];
26 } test_data_msg_t;
27
28 #endif // TEST_H
```

Test.h

TinyOS 기반 NesC 프로그래밍 (온습도센서)

- vim TestC.nc

```
~/sonnonet_tinyos/Tinyos/TestLowOneHopSht_sc
1 module TestC
2 {
3   uses {
4     interface Boot;
5     interface Leds;
6     interface Timer<TMilli> as MilliTTimer;
7
8     interface SplitControl as RadioControl;
9     interface AMSend as RadioSend;
10
11     interface Read<uint16_t> as Temp;
12     interface Read<uint16_t> as Humi;
13     interface Read<uint16_t> as Illu;
14
15     interface Battery;
16   }
17 }
18
```

TestC.nc

TestC.nc

```
~/sonnonet_tinyos/Tinyos/TestLowOneHopSht_sc
19 implementation
20 {
21     message_t testMsgBfrr;
22     test_data_msg_t *testMsg;
23
24     uint32_t seqNo;
25     uint8_t step;
26
27
28     task void startTimer();
29     event void Boot.booted() {
30         testMsg = (test_data_msg_t *)call RadioSend.getPayload(
31             &testMsgBfrr, sizeof(test_data_msg_t));
32         testMsg->srcID = TOS_NODE_ID;
33
34         seqNo = 0;
35
36         post startTimer();
37     }
38
39     task void startTimer() {
40         call MilliTimer.startPeriodic(TEST_PERIOD);
41     }
42
43     task void radioOn();
44     event void MilliTimer.fired() {
45         post radioOn();
46     }
47 }
```

TestC.nc

```
47
48 void startDone();
49 task void radioOn() {
50     if (call RadioControl.start() != SUCCESS) startDone();
51 }
52
53 event void RadioControl.startDone(error_t error) {
54     startDone();
55 }
56
57 task void readTask();
58 void startDone() {
59     step = 0;
60     post readTask();
61     call Leds.led0Toggle();
62 }
63
64 void sendDone();
65 task void sendTask() {
66     testMsg->seqNo = seqNo++;
67     testMsg->type = 2; //THL type 2
68
69     if (call RadioSend.send(AM_BROADCAST_ADDR, &testMsgBfrr,
70         sizeof(test_data_msg_t)) != SUCCESS) sendDone();
71     call Leds.led2Toggle();
72 }
73
```

TestC.nc

```
70 |
74 | event void RadioSend.sendDone(message_t* msg, error_t error) {
75 |     sendDone();
76 | }
77 |
78 | task void radioOff();
79 | void sendDone() {
80 |     call Leds.led00ff();
81 |     call Leds.led10ff();
82 |     call Leds.led20ff();
83 |     post radioOff();
84 | }
85 |
86 | void stopDone();
87 | task void radioOff() {
88 |     if (call RadioControl.stop() != SUCCESS) stopDone();
89 | }
90 |
91 | event void RadioControl.stopDone(error_t error) {
92 |     stopDone();
93 | }
94 |
95 | void stopDone() {
96 | }
97 |
```

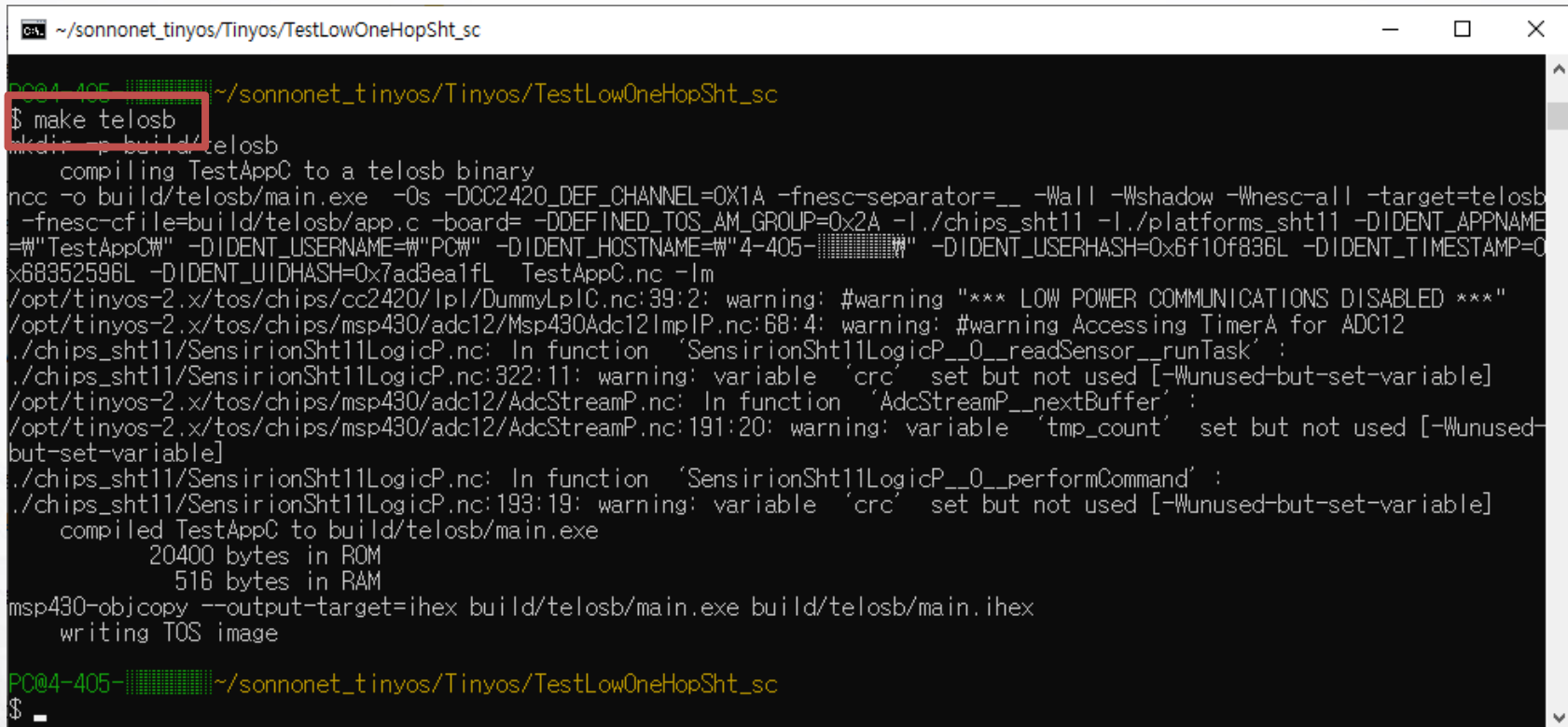
TestC.nc

```
97  task void readTask() {
98      switch(step) {
99          case 0:
100              call Temp.read(); break;
101          case 1:
102              call Humi.read(); break;
103          case 2:
104              call Illu.read(); break;
105          default:
106              testMsg->battery = call Battery.getVoltage();
107              post sendTask();
108              break;
109      }
110      step += 1;
111  }
112
113  event void Temp.readDone(error_t error, uint16_t val) {
114      //if (error != SUCCESS) call Leds.led00n();
115      testMsg->Temp = error == SUCCESS ? val : 0xFFFA;
116      post readTask();
117  }
118  event void Humi.readDone(error_t error, uint16_t val) {
119      //if (error != SUCCESS) call Leds.led10n();
120      testMsg->Humi = error == SUCCESS ? val : 0xFFFB;
121      post readTask();
122  }
123  event void Illu.readDone(error_t error, uint16_t val){
```

TestC.nc

```
123 event void lllu.readDone(error_t error, uint16_t val){  
124     testMsg->lllu = error == SUCCESS ? val : 0xFFFC;  
125     post readTask();  
126 }  
127 }
```

- 컴파일
 - make telosb



```
~/sonnonet_tinyos/Tinyos/TestLowOneHopSht_sc
$ make telosb
mkdir -p build/telosb
compiling TestAppC to a telosb binary
ncc -o build/telosb/main.exe -Os -DCC2420_DEF_CHANNEL=0x1A -fnesc-separator=__ -Wall -Wshadow -Wnesc-all -target=telosb
-fnesc-cfile=build/telosb/app.c -board= -DDEFINED_TOS_AM_GROUP=0x2A -I./chips_sht11 -I./platforms_sht11 -DIDENT_APPNAME
="TestAppC" -DIDENT_USERNAME="PCW" -DIDENT_HOSTNAME="4-405-1111111111" -DIDENT_USERHASH=0x6f10f836L -DIDENT_TIMESTAMP=0
x68352596L -DIDENT_UIDHASH=0x7ad3ea1fL TestAppC.nc -lm
/opt/tinyos-2.x/tos/chips/cc2420/lpl/DummyLpIC.nc:39:2: warning: #warning "*** LOW POWER COMMUNICATIONS DISABLED ***"
/opt/tinyos-2.x/tos/chips/msp430/adc12/Msp430Adc12ImpI.nc:68:4: warning: #warning Accessing TimerA for ADC12
./chips_sht11/SensirionSht11LogicP.nc: In function 'SensirionSht11LogicP__0__readSensor__runTask':
./chips_sht11/SensirionSht11LogicP.nc:322:11: warning: variable 'crc' set but not used [-Wunused-but-set-variable]
/opt/tinyos-2.x/tos/chips/msp430/adc12/AdcStreamP.nc: In function 'AdcStreamP__nextBuffer':
/opt/tinyos-2.x/tos/chips/msp430/adc12/AdcStreamP.nc:191:20: warning: variable 'tmp_count' set but not used [-Wunused-
but-set-variable]
./chips_sht11/SensirionSht11LogicP.nc: In function 'SensirionSht11LogicP__0__performCommand':
./chips_sht11/SensirionSht11LogicP.nc:193:19: warning: variable 'crc' set but not used [-Wunused-but-set-variable]
compiled TestAppC to build/telosb/main.exe
20400 bytes in ROM
516 bytes in RAM
msp430-objcopy --output-target=ihex build/telosb/main.exe build/telosb/main.ihex
writing TOS image
PC@4-405-1111111111~/sonnonet_tinyos/Tinyos/TestLowOneHopSht_sc
$
```


TinyOS 기반 NesC 프로그래밍 (온습도센서)

- Upload Fusing (kmote usb 연결 후)
 - make telosb install.(노드아이디)



```
~/sonnonet_tinyos/Tinyos/TestLowOneHopSht_sc
$ make telosb install.100
mkdir -p build/telosb
compiling testapp to a telosb binary
ncc -o build/telosb/main.exe -Os -DCC2420_DEF_CHANNEL=0x1A -fnesc-separator=__ -Wall -Wshadow -Wnesc-all -target=telosb
-fnesc-cfile=build/telosb/app.c -board= -DDEFINED_TOS_AM_GROUP=0x2A -I./chips_sht11 -I./platforms_sht11 -DIDENT_APPNAM
-W"TestAppCW" -DIDENT_USERNAME="PCW" -DIDENT_HOSTNAME="4-405-1111111111" -DIDENT_USERHASH=0x6f10f836L -DIDENT_TIMESTAMP=
x68352663L -DIDENT_UIDHASH=0x14254cd5L TestAppC.nc -lm
/opt/tinyos-2.x/tos/chips/cc2420/lpl/DummyLplC.nc:39:2: warning: #warning "*** LOW POWER COMMUNICATIONS DISABLED ***"
/opt/tinyos-2.x/tos/chips/msp430/adc12/Msp430Adc12ImpIP.nc:68:4: warning: #warning Accessing TimerA for ADC12
./chips_sht11/SensirionSht11LogicP.nc: In function 'SensirionSht11LogicP_0_readSensor__runTask':
./chips_sht11/SensirionSht11LogicP.nc:322:11: warning: variable 'crc' set but not used [-Wunused-but-set-variable]
/opt/tinyos-2.x/tos/chips/msp430/adc12/AdcStreamP.nc: In function 'AdcStreamP__nextBuffer':
/opt/tinyos-2.x/tos/chips/msp430/adc12/AdcStreamP.nc:191:20: warning: variable 'tmp_count' set but not used [-Wunused-
but-set-variable]
./chips_sht11/SensirionSht11LogicP.nc: In function 'SensirionSht11LogicP_0_performCommand':
./chips_sht11/SensirionSht11LogicP.nc:193:19: warning: variable 'crc' set but not used [-Wunused-but-set-variable]
compiled TestAppC to build/telosb/main.exe
 20400 bytes in ROM
   516 bytes in RAM
msp430-objcopy --output-target=ihex build/telosb/main.exe build/telosb/main.ihex
writing TOS image
tos-set-symbols --objcopy msp430-objcopy --objdump msp430-objdump --target ihex build/telosb/main.ihex build/telosb/main
.ihex.out-100 TOS_NODE_ID=100 ActiveMessageAddressC__addr=100
found mote on COM32 (using bsl:auto)
installing telosb binary using bsl
tos-bsl --telosb -c 31 -r -e -l -p build/telosb/main.ihex.out-100
MSP430 Bootstrap Loader Version: 1.39-telos-B
Mass Erase...
Transmit default password ...
Invoking BSL...
Transmit default password ...
Current bootstrap loader version: 1.61 (Device ID: f16c)
Changing baudrate to 38400 ...
Program ...
20904 bytes programmed.
Reset device ...
rm -f build/telosb/main.exe.out-100 build/telosb/main.ihex.out-100

PC@4-405-1111111111 ~/sonnonet_tinyos/Tinyos/TestLowOneHopSht_sc
$
```

oscilloscope.py

~/sono_raspberry/BaseStation_v1.1

```
1 # Data Format
2 # C02 Data0
3 # THL Temperature Data0, Humidity Data1, Illumination Data2, Battery Data3
4
5 import sys
6 import tos
7 import datetime
8 import threading
9
10 AM_OSCILLOSCOPE = 0x93
11
12 class OscilloscopeMsg(tos.Packet):
13     def __init__(self, packet = None):
14         tos.Packet.__init__(self,
15                             [
16                                 ('srcID', 'int', 2),
17                                 ('seqNo', 'int', 4),
18                                 ('type', 'int', 2),
19                                 ('Data0', 'int', 2),
20                                 ('Data1', 'int', 2),
21                                 ('Data2', 'int', 1),
22                                 ('Data3', 'int', 1),
23                                 ('Data4', 'int', 2),
24                             ],
25                             packet)
26
27 if '-h' in sys.argv:
28     print "Usage:", sys.argv[0], "serial@/dev/ttyUSB0:57600"
29     sys.exit()
```

oscilloscope.py

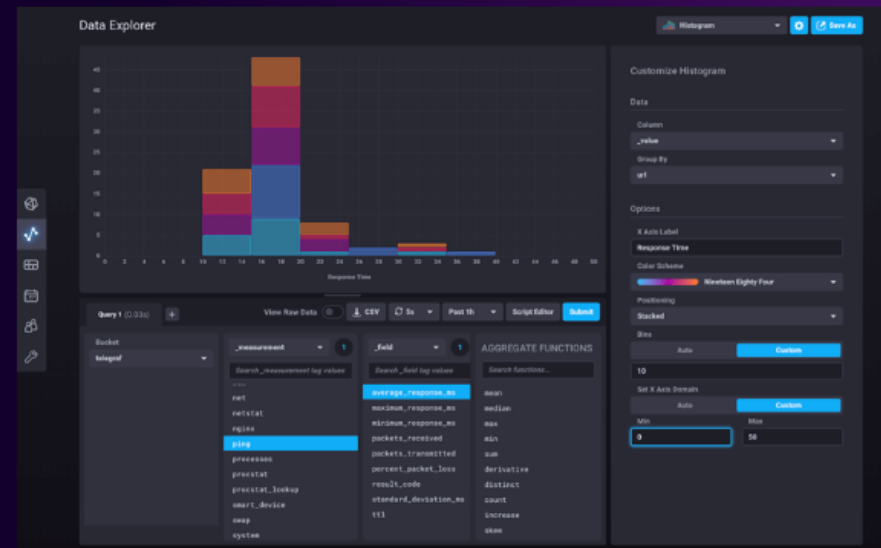
```
29 am = tos.AM()
30
31 while True:
32     p = am.read()
33     msg = OscilloscopeMsg(p.data)
34     print p
35
36 ##### THL Logic #####
37     if msg.type == 2:
38         battery = msg.Data4
39
40         Illumi = int(msg.Data2) + int(msg.Data3*256)
41         Illumi = Illumi
42         humi = -2.0468 + (0.0367*msg.Data1) + (-1.5955*0.000001)*msg.Data1*msg.Data1
43         temp = -(39.6) + (msg.Data0 * 0.01)
44         try:
45             with conn.cursor() as curs:
46                 Now = datetime.datetime.now()
47                 sql = """insert into JB_Sensor_THL(NODE_ID,SEQ,TEMPERATURE,HUMIDITY,ILLUMINATION,REGDATE)
48
49                 values(%s, %s, %s, %s, %s, %s)"""
50                 curs.execute(sql, (msg.srcID, msg.seqNo, temp, humi, Illumi, Now))
51                 conn.commit()
52         except all,e:
53             print e.args
54             conn.close()
55     print "id:" , msg.srcID, " Count : ", msg.seqNo, "
    "Temperature: ", temp, "Humidity: ", humi, "Illumination: ", Illumi, "Battery : ", battery
```

InfluxDB

InfluxDB

InfluxDB is a time series database designed to handle high write and query loads.

Get InfluxDB



InfluxDB 1.x

InfluxDB 1.x is the open source time series database component of the TICK Stack [Telegraf, InfluxDB, Chronograf, Kapacitor].

InfluxDB 2.0

Currently in beta, InfluxDB 2.0 incorporates everything you need in a time series platform into a single binary.

InfluxDB Cloud

InfluxDB Cloud is a fast, elastic, serverless time series platform as a service – easy to use with usage-based pricing

시계열데이터베이스 설치

InfluxDB 설치

- https://github.com/sonnonet/2024_inhatec

InfluxDB 설치

- InfluxDB download key using wget

```
wget -q https://repos.influxdata.com/influxdata-archive_compat.key  
echo '393e8779c89ac8d958f81f942f9ad7fb82a25e133faddaf92e15b16e6ac9ce4c influxdata-archive_compat.key' |  
echo 'deb [signed-by=/etc/apt/trusted.gpg.d/influxdata-archive_compat.gpg] https://repos.influxdata.com'
```

- Packages are up to date && install Influxdb

```
sudo apt-get update && sudo apt-get install influxdb -y
```

- InfluxDB as a background service on startup

```
sudo service influxdb start
```

- InfluxDB is status (service)

```
sudo service influxdb status
```

InfluxDB 데이터베이스 만들기

```
$ influx
```

```
>create database <데이터베이스이름>
```

```
확인 : show databases
```

시계열데이터베이스 설치

Python InfluxDB lib 설치

- https://github.com/sonnonet/2024_inhatc

influxdb import with python

```
pip install influxdb
```



Grafana 설치

- https://github.com/sonnonet/2024_inhatc

🔗 Grafana Installation

1. Install the prerequisite packages

```
sudo apt-get install -y apt-transport-https software-properties-common wget
```

2. Import the GPG key:

```
sudo mkdir -p /etc/apt/keyrings/  
wget -q -O - https://apt.grafana.com/gpg.key | gpg --dearmor | sudo tee /etc/apt/keyrings/grafana.gpg >
```

3. To add a repository for stable releases, run the following command:

```
echo "deb [signed-by=/etc/apt/keyrings/grafana.gpg] https://apt.grafana.com stable main" | sudo tee -a
```

4. Run the following command to update the list of available packages:

```
sudo apt-get update && sudo apt-get install grafana -y
```

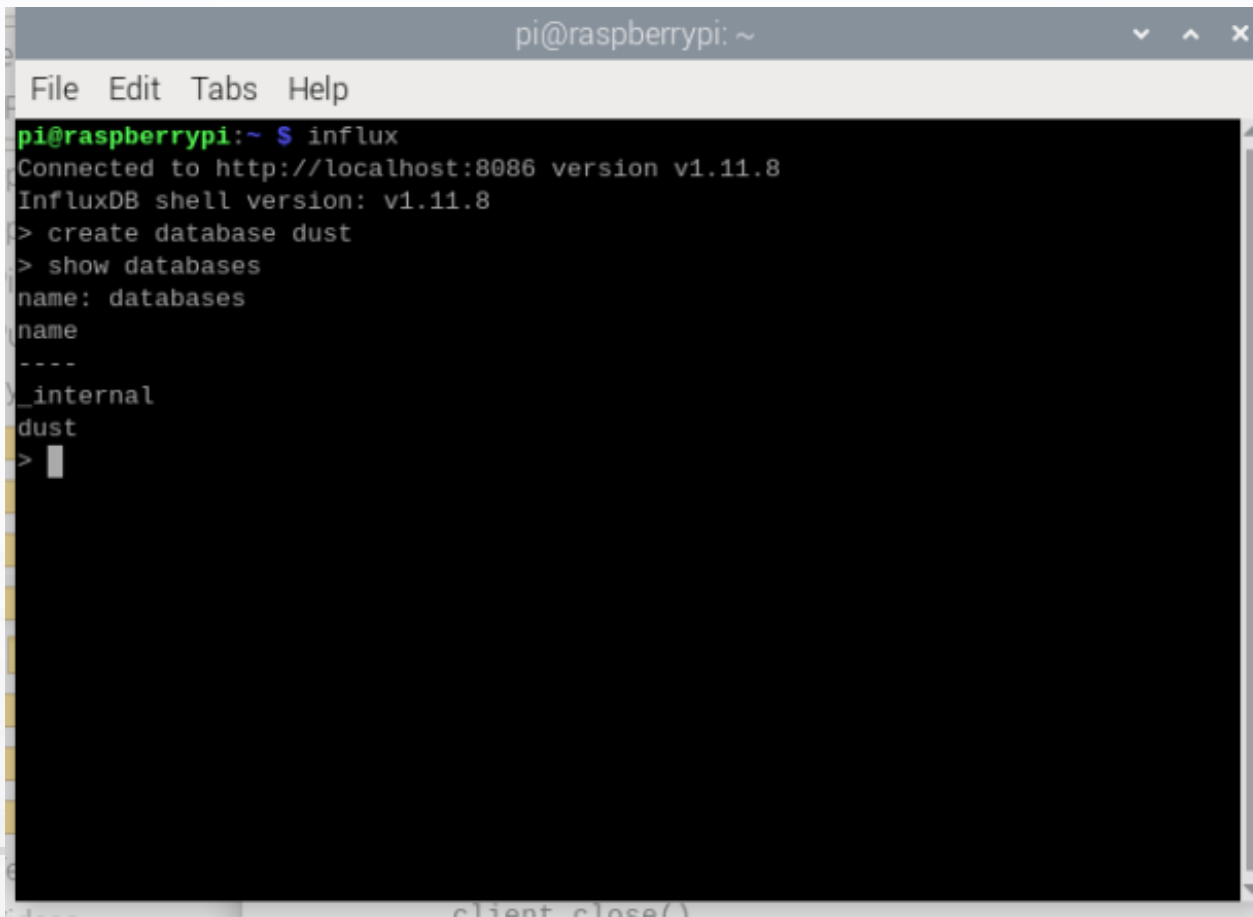
5. Run the following command to server start

```
sudo systemctl start grafana-server
```

오픈 소스 GUI 설치

influxDB Database 생성

```
$ influx  
$ create database dust  
$ show databases  
$ exit
```



```
pi@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~ $ influx  
Connected to http://localhost:8086 version v1.11.8  
InfluxDB shell version: v1.11.8  
> create database dust  
> show databases  
name: databases  
name  
----  
_internal  
dust  
> |
```


Arduino to Python Serial Middleware

\$ vim dustInfluxdb.py

```
1
2
3 import time
4 import requests, json
5 from influxdb import InfluxDBClient as influxdb
6 import serial
7
8 seri = serial.Serial('/dev/ttyACM0', baudrate = 9600, timeout = None)
9
10
11 while(True):
12     time.sleep(1)
13     if seri.in_waiting !=0:
14         content = seri.readline()
15         a = float(content.decode())
16         data = [{
17             'measurement' : 'dust',
18             'tags':{
19                 'InhaUni' : '2222',
20             },
21             'fields':{
22                 'dust' : a,
23             }
24         }]
```

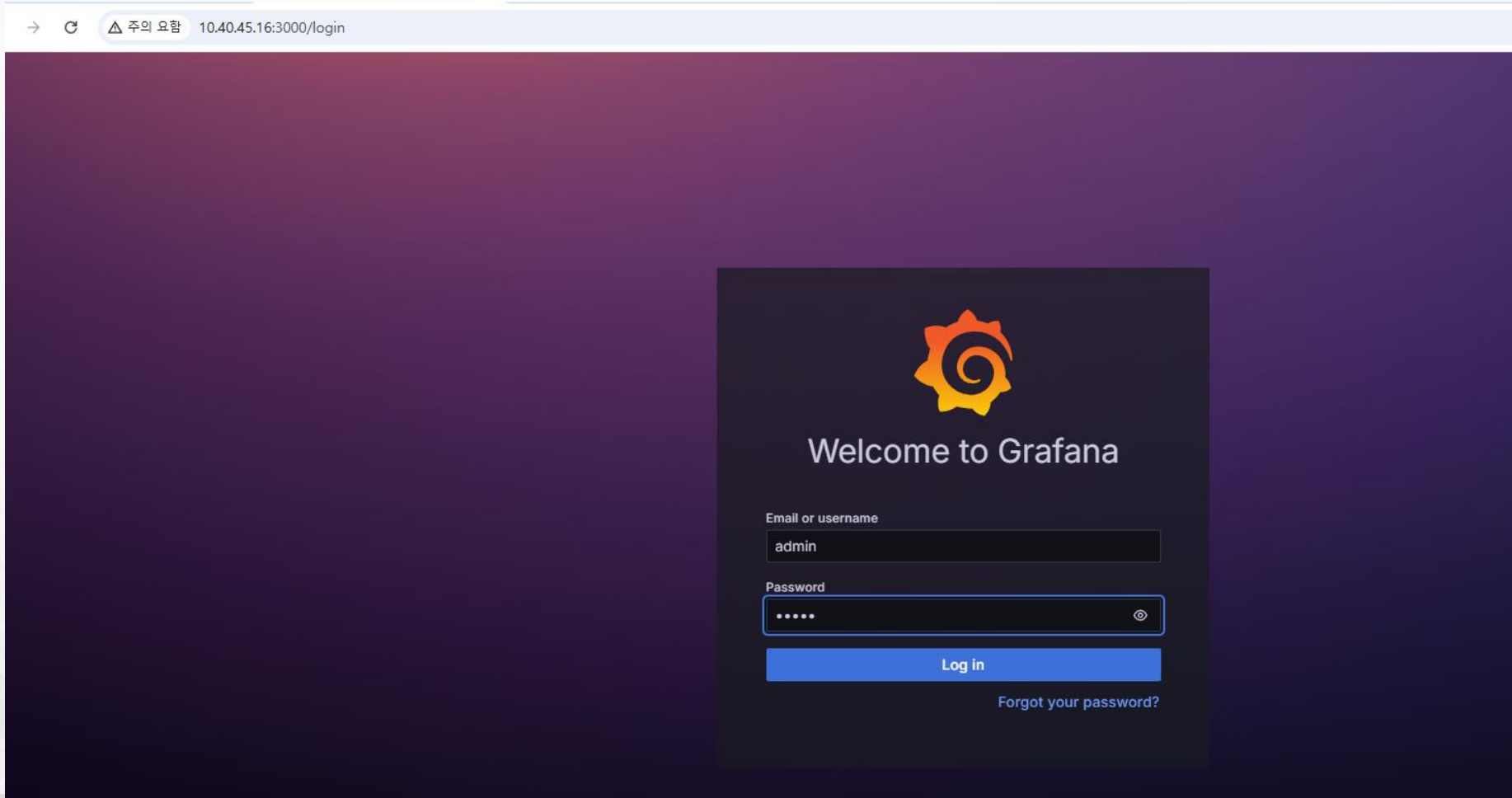
Arduino to Python Serial Middleware - 1

```
25     client = None
26     try:
27         client = influxdb('localhost',8086,'root','root','dust')
28     except Exception as e:
29         print ("Exception" + str(e))
30     if client is not None:
31         try:
32             client.write_points(data)
33         except Exception as e:
34             print("Exception write " + str(e))
35         finally:
36             client.close()
37     print(a)
38     print("running influxdb OK")
~
~
```

오픈 소스 GUI 설치

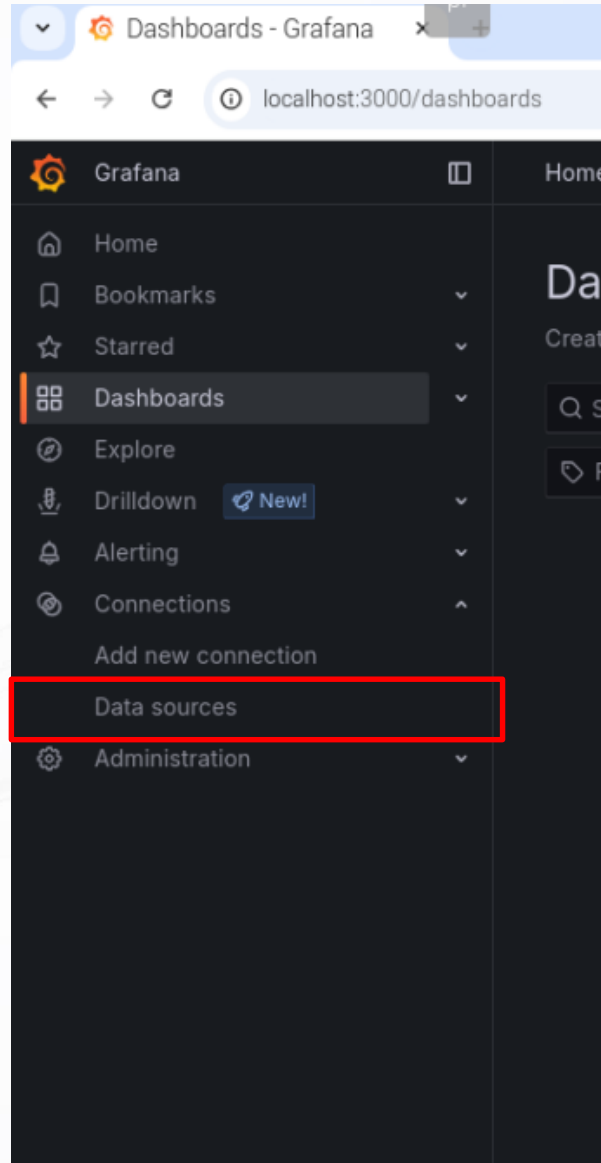
Grafana 접속

- 크롬미니 -> localhost:3000
- 기본 -> username : [admin](#) , password : [admin](#)



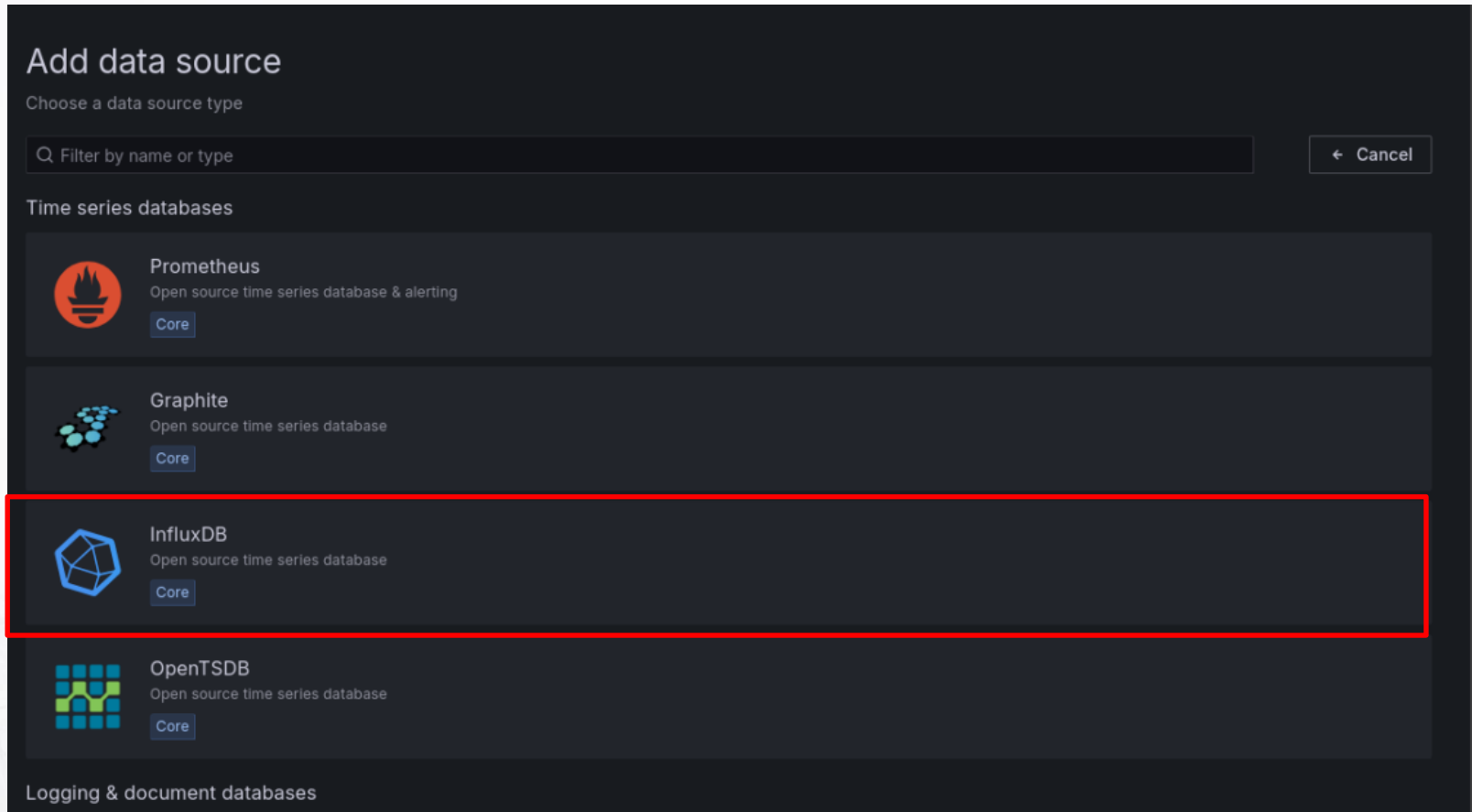
Grafana 설정

- Connections -> Data sources



Grafana 설정

- InfluxDB



오픈 소스 GUI 설치

Grafana 설정

- InfluxDB 연결 설정
- `http://localhost:8086`

Query language

InfluxQL

Please report any issues to:
<https://github.com/grafana/grafana/issues>

HTTP

URL

Your access method is `Server`, this means the URL needs to be accessible from the grafana backend/server.

`http://localhost:8086`

Allowed cookies

Grafana proxy deletes forwarded cookies by default. Specify cookies by name that should be forwarded to the data source.

New tag (enter key to add) Add

Timeout

HTTP request timeout in seconds

Timeout in seconds

Auth

Basic auth	☐	With Credentials	☒
TLS Client Auth	☐	With CA Cert	☒
Skip TLS Verify	☐		
Forward OAuth Identity	☐		

Custom HTTP Headers

Grafana 설정

- InfluxDB 연결 설정
- Database : dust
- User : root, password: root

InfluxDB Details

Database Access

Setting the database for this datasource does not deny access to other databases. T example: `SHOW MEASUREMENTS ON _internal` or `SELECT * FROM "_internal".."databa`

To support data isolation and security, make sure appropriate permissions are config

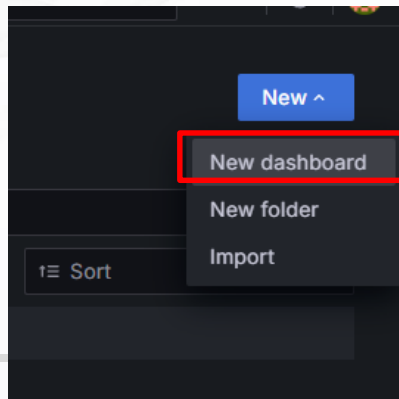
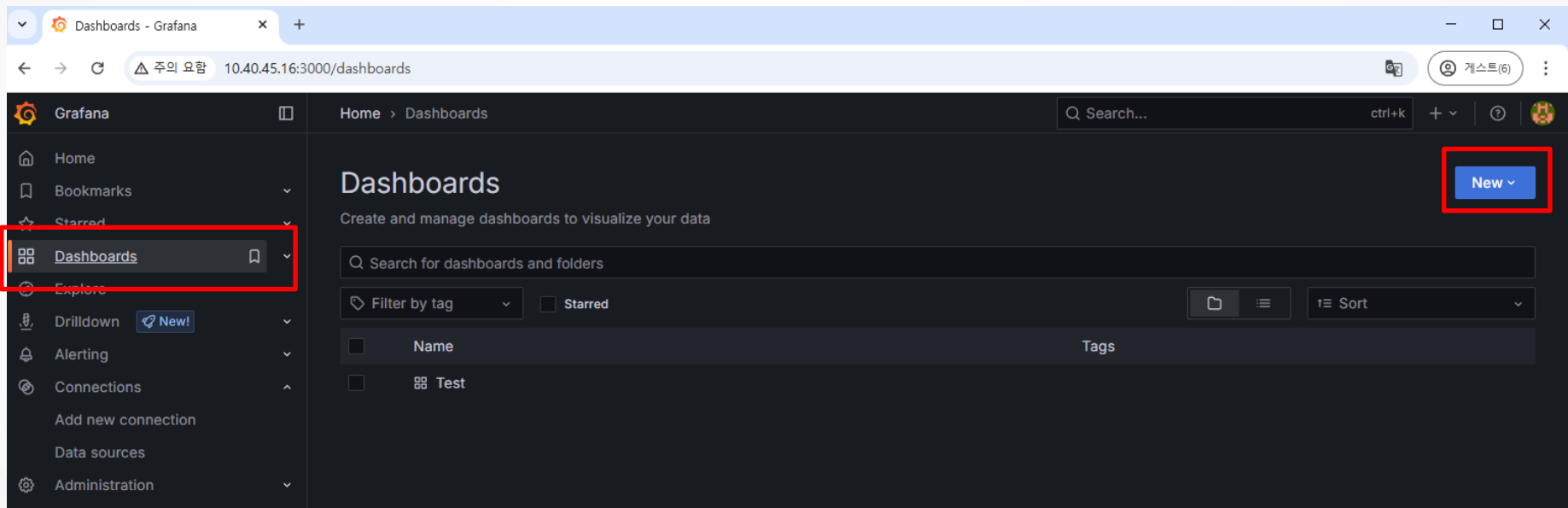
Database	dust
User	root
Password	
HTTP Method	Choose
Min time Interval	10s
Max series	1000

Delete Save & test

오픈 소스 GUI 설치

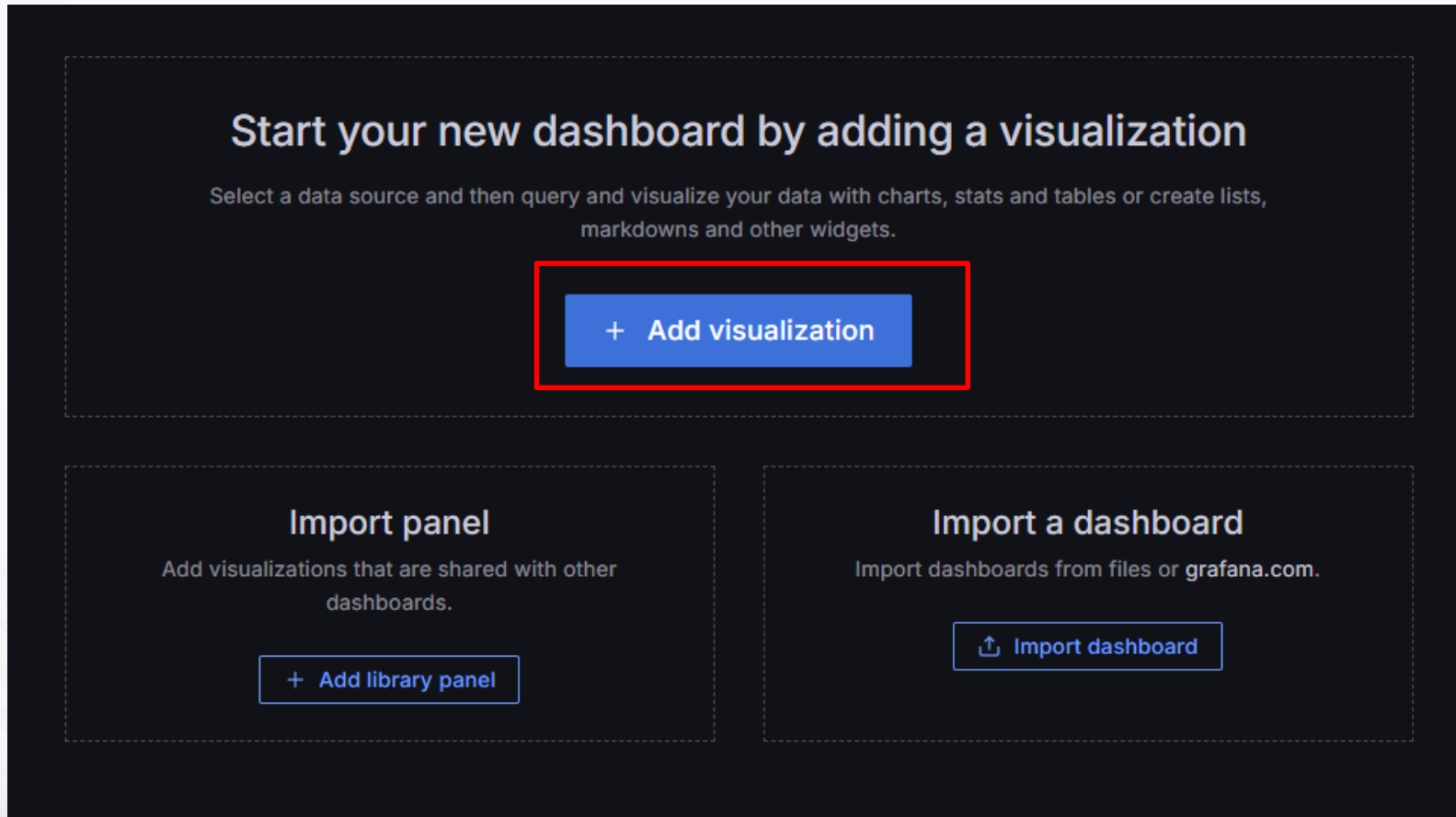
Grafana 설정

- Dashboards 설정



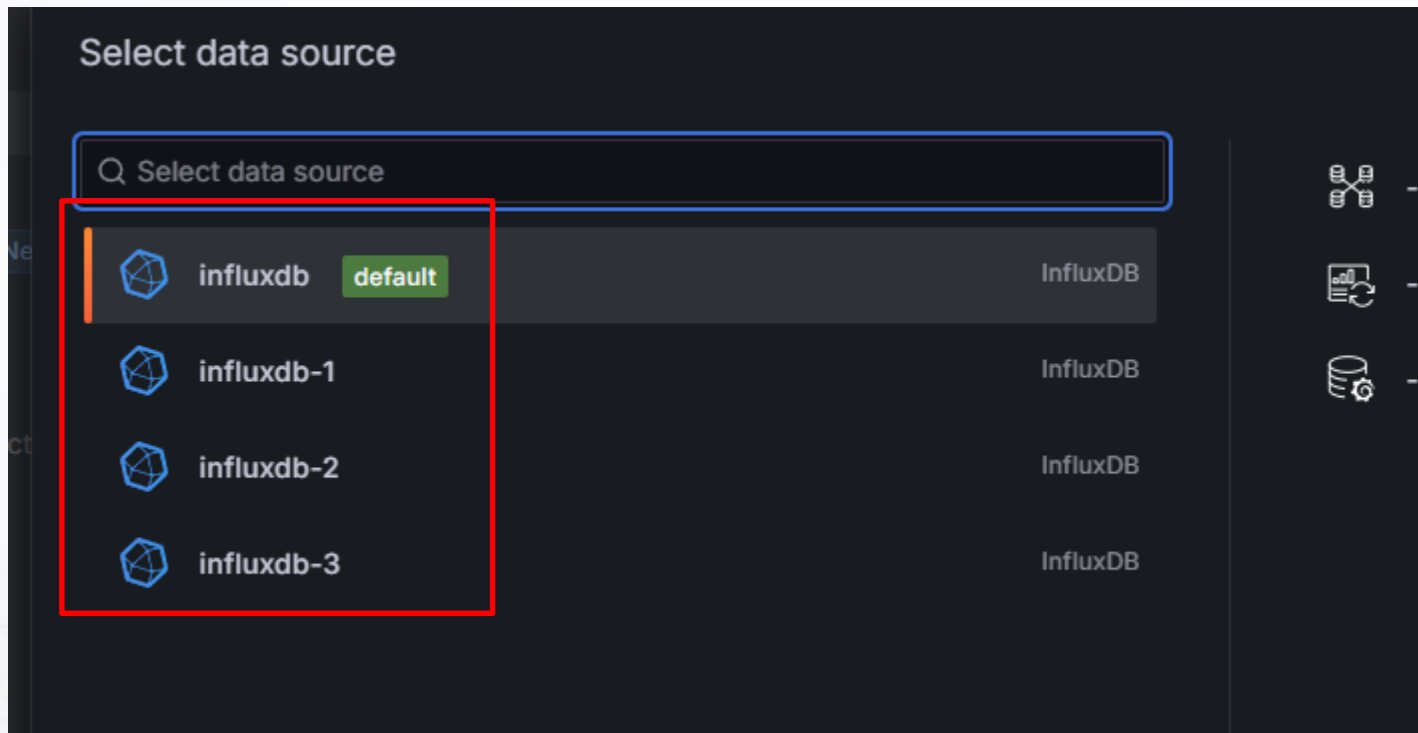
Grafana 설정

- Dashboards 설정 - 1



Grafana 설정

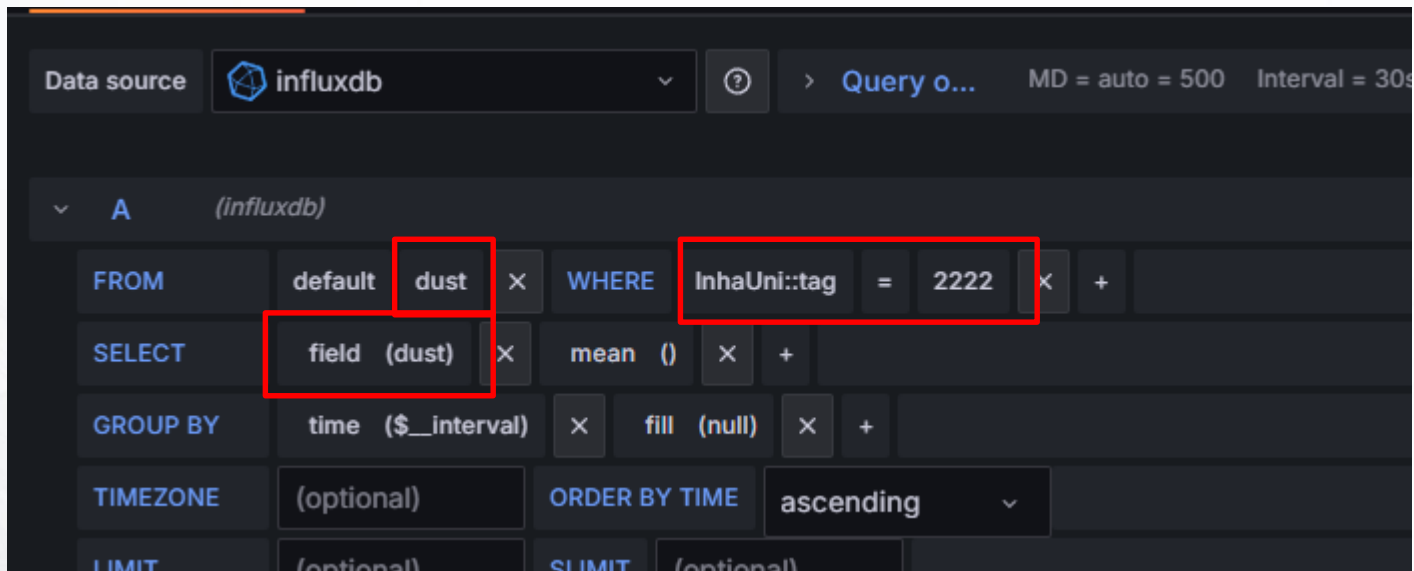
- Dashboards 설정 – 2
- 본인이 설정한 data source 지정



오픈 소스 GUI 설치

Grafana 설정

- Dashboards 설정 – 2
- select measurement -> dust
- tag -> inhaUni = 2222
- field(value) -> dust



Arduino to Python Serial using TelegramBot

- handler 등록

```
def main() -> None:
    """Run bot."""
    # Create the Application and pass it your bot's token.
    application = Application.builder().token("1331886552:

    # on different commands - answer in Telegram
    application.add_handler(CommandHandler(["start", "help"], start))
    application.add_handler(CommandHandler("set", set_timer))
    application.add_handler(CommandHandler("dust", dustToInfluxDB))
    application.add_handler(CommandHandler("unset", unset))
```

Arduino to Python Serial using TelegramBot

- 함수 구현

```
import cv2
import time
from telegram import Update
from telegram.ext import Application, CommandHandler, ContextTypes
from influxdb import InfluxDBClient as influxdb
import serial

seri = serial.Serial('/dev/ttyACM0', baudrate = 9600, timeout = None)

# Enable logging
logging.basicConfig(
    format="%(asctime)s - %(name)s - %(levelname)s - %(message)s", level=logging.INFO
)

async def dustToInfluxDB(context: ContextTypes.DEFAULT_TYPE) -> None:
    if seri.in_waiting != 0:
        content = seri.readline()
        a = float(content.decode())
        job = context.job
        await context.bot.send_message(job.chat_id, text=f"Dust Sensor Value! {a}")

# Define a few command handlers. These usually take the two arguments update and
# context.
# Best practice would be to replace context with an underscore,
```

Arduino to Python Serial using TelegramBot

- set_timer 수정

```
async def set_timer(update: Update, context: ContextTypes.DEFAULT_TYPE) -> None:
    """Add a job to the queue."""
    chat_id = update.effective_message.chat_id
    try:
        # args[0] should contain the time for the timer in seconds
        due = float(context.args[0])
        if due < 0:
            await update.effective_message.reply_text("Sorry we can not go back to future!")
            return

        job_removed = remove_job_if_exists(str(chat_id), context)
        context.job_queue.run_repeating(dustToInfluxDB, due, chat_id=chat_id, name=str(chat_id), data=due)

        text = "Timer successfully set!"
        if job_removed:
            text += " Old one was removed."
        await update.effective_message.reply_text(text)
```

TinyOS 기반 NesC 프로그래밍 (환경구축)

- 라즈베리파이 기반 TinyOS 환경 구축

Tinyos 설치 (2023-12)

- 라즈베리파이 3B+
- 사전 설치

```
$ sudo apt-get install emacs gperf bison flex git build-essential automake avarice avr-libc msp430-libc avrdude binutils-avr binutils-avr
```

- 사전 설정

```
$ vim ~/tinyos.env

export TOSROOT=/opt/tinyos-2.x
export TOSDIR=$TOSROOT/tos
export CLASSPATH=$TOSROOT/support/sdk/java/tinyos.jar:$CLASSPATH
export CLASSPATH=$TOSROOT/support/sdk/java:$CLASSPATH
export CLASSPATH=.:$CLASSPATH
export MAKERULES=$TOSROOT/support/make/Makerules
export PYTHONPATH=$PYTHONPATH:$TOSROOT/support/sdk/python

echo "setting up TinyOS on source path $TOSROOT"
```

```
$ vim ~/.bashrc
글쎄에 추가
source ~/tinyos.env
```

```
$ source ~/.bashrc
```

TinyOS 기반 NesC 프로그래밍 (환경구축)

- JNI 파일 복사

```
$ git clone https://sonnonet/jni.git  
$ cd sonnonet/jni/tinyos/  
$ sudo mv libtoscomm.so /usr/lib  
$ sudo mv libgetenv.so /usr/lib
```



- NesC 설치

```
$ mv ~/sonnonet/jni/tinyos/nesc-1.3.4.tar.gz ~/   
$ tar -xvzf nesc-1.3.4.tar.gz  
$ cd nesc-1.3.4  
$ ./configure  
$ make  
$ sudo make install
```



TinyOS 기반 NesC 프로그래밍 (환경구축)

- TinyOS-Tools 설치

```
$ cd /home/pi
$ git clone --branch release_tinyos_2_1_2 https://github.com/tinyos/tinyos-release.git
$ mv tinyos-release tinyos-2.x
$ sudo mv tinyos-2.x /opt/
```

```
$ cd /opt/tinyos-2.x/tools/
$ ./Bootstrap
$ ./configure
$ make
$ sudo make install
```

- 테스트

```
$ motelist
```

- 에러

```
attributeerror 'Serial' object has no attribute 'setbaudrate'
-> sudo vim /usr/local/bin/tos-bsl
-> :1345 줄트 이동
-> self.serialport.setBaudrate(baudrate) -> self.serialport.setBaudrate = baudrate (수정)
```

12~13주차 수업이 끝났습니다

고생하셨습니다.

